



DHMFRD – TER: a deep hybrid model for fake review detection incorporating review texts, emotions, and ratings

Ramadhani Ally Duma^{1,2} · Zhendong Niu¹ · Ally Nyamawe² · Jude Tchaye-Kondi¹ · James Chambua³ · Abdulganiyu Abdu Yusuf¹

Received: 16 June 2022 / Revised: 3 February 2023 / Accepted: 30 March 2023 /

Published online: 26 May 2023

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Abstract

Recently, there has been an increasing reward to manipulate product/ service reviews, mostly profit-driven, since positive reviews infer high business returns and vice versa. To combat this issue, experts in industry and researchers recently attempted integrating multi-aspect (reviewer- and review-centric) data features. However, the emotions hidden in the review, the semantic meaning of the review, and data heterogeneity still deserve more study as they are essential indicators of fake content. This study proposed a Deep Hybrid Model for Fake Review Detection incorporating review Texts, Emotions, and Ratings (DHMFRD – TER). Initially, it computes contextualized review text vectors and extraction of emotion indicators representations. Then, the model learns the representation to extract higher-level review features. Finally, contextualized word vectors, ratings, and emotions are concatenated; such a multidimensional feature representation is used to classify reviews. Extensive experiments on three publicly available datasets demonstrate that DHMFRD–TER significantly outperforms state-of-the-art baseline approaches, achieving an accuracy of 0.988, 0.987, and 0.994 in Amazon, Yelp CHI, and OSF datasets, respectively.

Keywords Fake reviews · Ratings · Emotions · Convolution neural network · Long short-term memory · Bidirectional Encoder Representations from Transformers

1 Introduction

The internet has facilitated the rapid growth in business information sharing, in which customers can share experiences and feedback on a specific product or service. Generally, e-business has grown dramatically, resulting in the establishment of numerous online forums. These online forum portals have amplified the availability of complete instant information for a given product, such as payment options, price comparisons, and flexibility (which can be accessed anytime, anywhere). Shared information includes product/

✉ Zhendong Niu
zniu@bit.edu.cn

Extended author information available on the last page of the article

service user ratings and reviews text in which customer share their positive or negative feelings on a particular product or service [1]. The review information benefits both customers and business entities [2]. While businesses get to identify issues related to their product, customers acquire experience about a particular product for better purchase decisions [3].

Reviews are being manipulated since they have a direct economic impact on both merchants and customers. Positive reviews imply high revenue and vice versa [2, 4, 5]. Fake reviews can sometimes be written by friends, family, bots, or employees of the company or competitor. They are simply reviews written by someone who has not actually used the product or service. Statistics show about 81% of customers who read available product or service information, 79% make the right decision [3, 6]. Additionally, Luca (2016) [7] reported that any rating increase by at least 1 star causes an increase in revenue from 5 to 9 percent. It is seen as an opportunity for companies to hire deceivers to manipulate reviews. Online reviews have become a significant concern today. Ellison (2018) [8] reported that every third review on TripAdvisor was found to be fake.

The deceptive reviews detection process can be either manually, automatically (using machine/ deep learning (ML/DL)), or both techniques [9]. Someone can do manual spam review detection by looking at language structure variation, use of brand names, and first-person pronouns (I, my). Research findings show that fake reviews contain more first-person pronouns and brand names [10, 11, 12]. However, manual identification is often inaccurate and expensive in terms of time and manpower compared to automatic approaches [11]. Over the last decade, scholars have devised several automatic spam review detection methods, utilizing text similarity features, overall review rating deviation, review content, product-specific information, the semantic meaning of reviews, and reviewer behaviors. However, due to the economic benefits attached to it, spammers are also trying hard to reduce the probability of being easily detected; sometimes, fabricated reviews are drafted to sound like genuine ones. Traditional methods cannot effectively detect spam reviews due to limited features, data representations, and manipulations [6, 13, 14].

Scholars believe fake clues are in review text and hence developed various deceptive identification methods such as psycho-linguistic features, bag-of-words (n-grams), part-of-speech tagging (POS), and review duplicates [15]. On the other hand, these methods were evaluated by various scholars. For example, the study by Dong et al. (2018) [16] deduced that implicit information is more critical in spam detection as compared to explicit information, which is often used. Spam reviews are the ones that deviate much in rating scores from the rest [17]. Ren and Ji (2017) [18] and Zhang and Wallace (2015) [19] explicitly stated that it is difficult to get the semantic meaning of a review by only using linguistic (text) features due to sparsity problems. In addition, relying only on the text aspect is not effective, as there is no significant difference between words in fake and benign reviews (specifically on real-world commercial websites) [20, 21, 22]. Furthermore, spam reviews typically have more emotionally charged words and fewer factual numbers [23, 24]. Since the prime spammer's intention is to either promote or demote the product/ service, they usually try to make their review look authentic (real), but difficult to hide their emotions.

Recently, the DL methods have proved to be more powerful in feature extraction, which makes them gain a reputation in various natural language processing (NLP) applications [25]. Deep learning methods are inevitable due to reviews' complex high-dimensional nature and spammers' behaviors [6, 26]. Despite scholars' significant efforts to overcome highlighted issues, they still suffer from inadequate performance, classifier convergence, sparsity, etc.

To address the above-highlighted issues, we propose a novel unified framework that jointly learns from review texts, emotions, and ratings, namely DHMFRD-TER, to classify deceptive reviews more accurately. This research aims to consider reviews' implicit and explicit syntactic contextual information. It should be noted that we adopted some techniques, such as text

preprocessing and feature extraction, and fusion mechanisms from our recent work [3]; this work, in particular, incorporated emotions which is an essential feature for review classification. First, review text input tokenized with the help of bidirectional encoder representations from transformers (BERT) tokenizer; the tokenized text was taken as input to the BERT embedding layer, which builds contextualized embedding vector values. Second, emotions were extracted using lexicon-based emotion indicators and then concatenated with text vector values. Third, a convolutional neural network (CNN) receives vector representations comprised of BERT output (encoded vectors) and emotion indicators for extracting high-dimensional feature maps. Fourth, long short-term memory (LSTM) receives spatial feature representations (of text and emotion indicators) to extract long-term contextual dependencies between tokens. In order to overcome gradient vanishing problems, we adopted a three-gated LSTM with memory cells to capture long-term data dependencies, in which the obtained representation is then concatenated with an overall rating aspect. Finally, a dense layer uses the representation to classify the review (non-fake or fake). For verifying the effectiveness of our model, various performance metrics were used and compared with its variants. The main contributions of this work are summarized as follows:

- We propose a novel multi-feature fusion mechanism named DHMFRD–TER that jointly integrates review text vector representations, emotion indicators, and rating aspects to accurately identify spam reviews.
- Extensive experiments are conducted on three publicly available datasets to analyze the effectiveness of the proposed DHMFRD–TER. It achieves superior performance and outperforms the state-of-the-art baseline approaches in terms of accuracy of 0.988, 0.987, and 0.994 in Amazon, Yelp CHI, and OSF datasets, respectively (see Table 3).

The remainder of this paper is organized as follows. In Sect. 2, we discussed related work on fake reviews. Section 3 describes the methodology and techniques used for this research work, including comparative analysis with previous studies. Section 4 outlines the experimental setup and empirical evaluation of results compared with state-of-the-art methods. Finally, Sect. 5 concludes the paper and proposes future directions in fake review detection.

2 Related works

The terms *spam review*, *fake review*, *deceptive review*, and *opinion spam* are equivalent and can be substituted for one another. Deceptive reviews fall into three types: *type one*, *type two*, and *type three* [6, 27, 28, 29]. *Type one* refers to all reviews intended to ruin the image of a competitor's product/ service while promoting others. *Type two* is affiliated only with a particular service provider or manufacturer. *Type three* is just advertisements with no opinion related to the product/ service. The first paper on product spam review detection context is believed to have been published in 2007 by Jindal and Liu (2007) [30]. In their analysis, they used about 2.14 million Amazon product reviews dataset. The detection was based on product features and review similarity, assuming spammers usually post or duplicate their reviews.

Inspired by spam review pioneers (i.e., Jindal and Liu (2007) [30]), there is a significant number of heuristic studies proposed to solve the problem. For instance, the studies by Ott et al. (2011) [31] and Li et al. (2014) [32] effectively used content of the review, parts of speech tagging (POS-tagging), n-gram, and linguistic inquiry and word count (LIWC) to detect fictitious opinion. However, they did not consider reviewer behavior, which is important in detecting deceptive opinions. Recently, neural networks (NNs) and deep NN, in general, have been

proven to be more effective compared to standard ML methods [18]. To overcome basic ML limitations, Hajek et al. (2020) [33] recommended deep NNs such as CNN, LSTM, and deep feed-forward neural networks (DFFNNs) to help in extracting reviews' complex nature, multi-dimensionality, and abstraction features. According to Zhao et al. (2018) [34], fake reviews are usually short; hence CNN is more appropriate for short spam review identification.

In another development, Li et al. (2021) [35] and Mukherjee et al. (2013) [36] presented a method to detect group deceivers by building a relational model among reviewers and reviewer groups. Fei et al. (2013) [37] and Li et al. (2017) [38] exploited review burstiness and co-bursting distribution, respectively, to identify reviews and spammers. Individual or group spammers may use different IDs to hoax different reviewers, but their reviews look to be similar [6]. Li et al. (2011) [39] and Lim et al. (2010) [40] used reviews cosine similarity in establishing their spam review detection system. Mukherjee et al. (2012) [41] addressed spamming activity using behavior features such as rating distribution, user profile, and length of the reviews. In addressing the cold-start problem in fake reviews detection, Wang et al. (2017) [22] used the CNN model to incorporate behavior and text features into review sentence representations. Melleng et al. (2019) [42] used combined text representations of sentiments and emotion indicators to identify spam reviews. Peng and Zhong (2014) [43] revealed that sentiment strength is more effective as compared to rating scores when detecting spam reviews. Apart from rating and sentiment strength, the use of review semantics is also an important aspect of review classification. Review semantics can be extracted through the word embedding technique, but extracted vectors for word representation cannot extract emotions attached to the reviews. In addition, lower classification performances were the major limitations of the aforementioned studies.

There are several recent works reported to have higher performance results. For example, Asghar et al. (2020) [44] suggested a hybrid classification approach incorporating opinion deceptive review and reviewer information. The study also presented a revised weighting, feature selection scheme, and spamicity scores computation. Noekhah et al. (2014) [45] proposed an unsupervised graph structure-based method that utilizes the relationship between the reviewer (spammer) and review features. Kokate and Tidke [46] employed the J48 classifier to identify fake movie reviews and brand spam. Rajamohana and Umamaheswari [47] introduced a hybrid-binary approach using the "binary particle swarm and shuffled frog leaping" algorithm to optimize and select high-dimensional review features. Jacob and Selvi Rajendran [48] used a FABC-based CNNLSTM and semantic feature to distinguish between fake and authentic reviews. However, their approaches were limited by small and balanced experimental datasets. Hajek et al. [33] also presented a CNN and DFFNN that integrated review word embedding vectors (n-gram) and emotional features. They employed four datasets that covered both balanced and imbalanced aspects but were still small for generalization. Our recent work [3] presented a new method for identifying fake reviews in online businesses, which jointly learns from latent text feature vectors, aspect ratings, and overall ratings. However, there are two limitations. Firstly, we only evaluated our approach on one dataset, which is unsuitable for generalization. Secondly, we did not take into account emotional features, which are crucial in detecting spam reviews.

In summary, regardless of the higher performance achieved by utilizing DL approaches, we observed the following; to extract the semantic meaning of reviews, most of the existing works have used non-contextual word embedding techniques such as word2vec, tf/ tf-idf, and GloVe, which are less effective due to the contextual based nature of reviews. We adopted the BERT not only because it is contextual in nature, but also proved recently to be a more powerful feature extraction technique in spam identification and NLP projects in general [49, 50, 51]. As explained earlier, it is easy for spammers to manipulate review text and rating scores compared to emotions. Still, few studies consider the power of emotion in spam review detection. Additionally, we consider the utilization of small and balanced experimental datasets as a significant

limitation. To address these issues, we propose a novel DHMFRD–TER to capture the complex nature of review representations and enable the integration, which jointly leverages textual semantic context, emotion indicators, and rating aspects for effective fake review detection. We demonstrate the effectiveness of our solution using the three publicly available datasets.

3 Methodology

In this part, we present and describe all important details of the proposed review spam detection model, including emotion feature extraction, review text processing, feature concatenation, and review classification.

3.1 Emotion and overall rating features

The used dataset’s content includes the review text and overall rating values (more details will be explained later in Sect. 4). Emotion words in the review can determine the reviewer’s positive, neutral, and negative feelings [52, 53]. To ensure the reliability of emotion analysis, we used the results of the combination of two lexicons, namely *emotion-polarity*- and *emotion-based* features, as described below. For emotion-polarity-based feature indicators, the OpinionFinder [54] and the Bing Liu opinion lexicon [55] were selected and used, in which polarity features were obtained by calculating the numbers of positive Pos_e , neutral Neu_e and negative Neg_e words matching lexicons (see algorithm 1).

Algorithm 1 Obtaining polarity-based feature from review text

Input: Review R_v , number of segments N_s , Lexicon L_x

Output: Polarity-based features representation of N_s

1. Split R_v in N_s identical segments $e_1, \dots, e_{N_s}$
2. **for all** $e_1, \dots, e_{N_s}$ **do** {
3. Tokenize e_i into a set of words, W
4. Retrieve polarity values for all W matching L_x
5. Compile all positive polarity values as Pos_e in W
6. Compile all neutral polarity values as Neu_e in W
7. Compile all negative polarity values as Neg_e in W
8. for all the segments S_g of R_v , $v_i = [Pos_e, Neu_e, Neg_e]$
9. }
10. **end for**
11. Concatenated vector of review $v(R_v) = [v_1 \oplus v_2 \oplus \dots \oplus v_{N_s}]$
12. **return** $v(R_v)$

The reason behind the selection of these lexicons is that they are the most reliable hand-crafted dictionary resources in which sentiment values were assigned by multiple human judgments [56]. In addition, compared to ML approaches, lexicon-based emotion is rarely susceptible to indirect indicators of possible generated fake sentiment patterns [33]. To obtain emotion-based features, we employed the NRC emotion lexicon in which eight basic emotions (joy, sadness, trust, surprise, anger, disgust, fear, and anticipation) were put into consideration based on Plutchik's wheel of emotions. The API takes review text R_v (input) for processing, then, it returns (output) category vector values v of emotion for each word matching the lexicon. Finally, v for all words matching the L_x were concatenated to obtain the entire review representation $v(R_v)$. Overall, we obtained 12 emotion indicators (4 polarity-based and 8 emotion-based features).

3.2 The DHMFRD–TER

As introduced earlier, we adopted text preprocessing, feature extraction, and fusion mechanisms from our recent work [3]. The proposed model architecture, as illustrated in Fig. 1, combines three different neural network architectures: (1) BERT that uses a transformer architecture to process review text inputs, allowing it to capture contextual word vectors effectively, (2) CNN component is used to extract local features of review, and (3) LSTM component is used to capture sequential information and maintain context over longer input sequences. The model operation is in a feed-forward fashion where it first passes the input text through the BERT component to obtain contextualized word embedding vectors. The obtained embedding vectors and emotions are fed through several one-dimension convolutional layers (i.e., Conv1D) in the CNN module to extract local features. The output of the CNN layer is then fed into the LSTM to capture sequential information and maintain context over longer input sequences. The output of the LSTM is concatenated with overall ratings. Finally, the concatenation result is passed to a dense for classification.

The CNN block layer receives vector representations comprised of BERT output (encoded vectors) and emotion indicators for extracting high-dimensional feature maps. We employed a rectified linear unit (ReLU) and dropouts in pairs of multiple sequential convolutional layers. Mathematically, the ReLU is expressed in Eq. 1, where R_o is output, i is input, and $i > 0$.

$$R_o = \max(0, i) \quad (1)$$

The dropout was employed during training to prevent overfitting. Ideally, it randomly drops neural network units during training [57]. Equation 2 describes a simple dropout process, where ε is anticipated results, ϑ is a related value for representation, and it drops whenever $\vartheta = 1$; otherwise, it activates for other values.

$$f(\varepsilon, \vartheta) = \begin{cases} \vartheta, & \text{if } \varepsilon = 1 \\ 1 - \vartheta, & \text{if } \varepsilon = 0 \end{cases} \quad (2)$$

3.2.1 Review text and emotional features representation

Review text features T_f : Feature maps (new feature extraction) r_j is obtained through convolution operation on n words embedding vectors $\bar{w}_{1:n}$ (see Eq. 3) where $\odot$ is a convolutional operator, β is bias. At a given window size w , and a vector word length l then the convolutional kernel $y_j \in K^{w \times l}$.

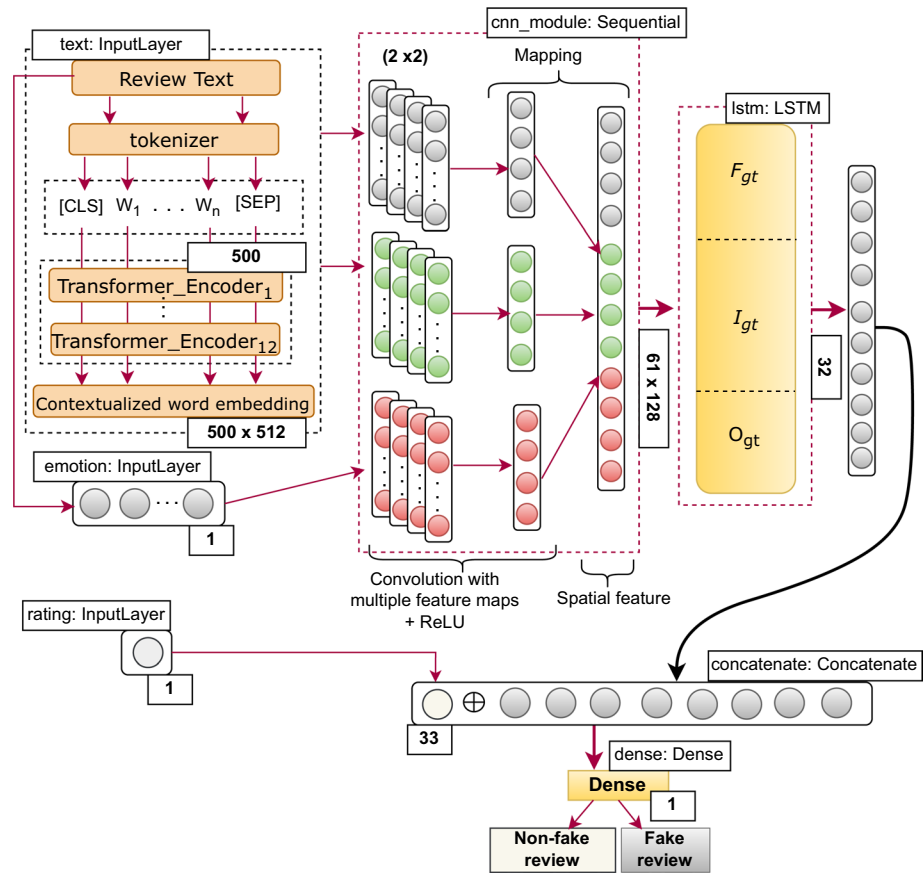


Fig. 1 Architecture of DHMFRD-TER

$$r_j = \text{ReLU}(\bar{w}_{1:j} \odot y_j + \beta_j) \quad (3)$$

When the filter convolves for the given w , starting from initial position j to end position $j - 1$, it concatenates/ maps a high-level feature window for each filter at a time (i.e., $\bar{w}_{1:j} = [y_j + y_{j+1} + \dots + y_{j-1}]$). After convolution, the pooling (max-pooling) operation helps to select important information only (i.e., removing low activation information) for each feature map. In this work, we adopt three convolution filters (x_1, x_2, x_3) on the word vectors in which we obtain three feature maps. To obtain an overall spatial text feature vector representation T_f , each window feature r_j is concatenated (see Eq. 4). The obtained T_f (the result of concatenation operation) is considered a latent feature representation of a review text.

$$T_f = [r_1 \oplus r_2 \oplus r_3] \quad (4)$$

Emotion features E_f : Let $e_i \in K^n$ be an n -dimensional vector of emotion indicators features, and $e_{i:i+j}$ be the concatenation of emotion indicator features $e_i, e_{i+1}, e_{i+2}, e_{i+3}, \dots$.

e_{i+j} . The convolutional filter matrix $f \in K^{n \times v}$ is applied to obtain new feature maps e_{ej} as expressed in Eq. 5

$$e_{ej} = \text{ReLU}(f_e \odot e_{1:n} + \beta_j) \quad (5)$$

Overall spatial emotion indicators feature representations E_f , each window feature e_j were concatenated as expressed in Eq. 6. Obtained E_f is considered as latent emotion indicators representation of a review text.

$$E_f = [e_1 \oplus e_2 \oplus e_3] \quad (6)$$

To obtain single vector feature map representations V^{t+e} we merged the outputs of Eqs. 4 and 6 as expressed in Eqs. 7, 8, and 9; where, W_t and W_e are the weight matrices, β_t , and β_e are respective biases.

$$T_f = \text{ReLU}(W_t r + \beta_t) \quad (7)$$

$$E_f = \text{ReLU}(W_e e + \beta_e) \quad (8)$$

$$V^{t+e} = T_f \oplus E_f \quad (9)$$

The LSTM retains long-term memory with the help of hidden units [58]. In this study, the LSTM receives an overall spatial feature representation V^{t+e} to extract long-term contextual dependencies between tokens in a sequential manner. We adopted a three-gated LSTM with the memory cell $\bar{S}_t$ (to capture the long-term data dependencies) in order to prevent the problem of exploding and gradient vanishing. Gates are used to setting the weights on the connections between the neural network and the $\bar{S}_t$ which also has its connections. When the forget gate F_{g_t} value is 1, connection weight is also 1. When the $\bar{S}_t$ retains the content with the value of 0 in F_{g_t} (the $\bar{S}_t$ discards its contents). When the input gate I_{g_t} is 1, the neural network writes into the $\bar{S}_t$, and when the output gate O_{g_t} is 1, the network can read the values from the $\bar{S}_t$ [3].

The output of the LSTM unit is then concatenated with overall ratings to obtain substantial data representation R_v^n for effective review classification. Let T^n and R be n vectors of review text (i.e., V^{t+e}) and an overall rating vector representation, respectively. The direct sum of $T^n \oplus R$ is a set of vectors (u, v) , where $u \in T^n$, $v \in R$. Then, $T^n \oplus R \mapsto R_v^n$. The corresponding vectors can be expressed as in Eq. 10

$$(u, v) \mapsto (u_1, u_2, \dots, u_n, v) \quad (10)$$

Finally, obtained R_v^n classified using the dense layer that condenses to 1-dimension output space. Generally, spam detection is usually treated as a binary classification problem in which a review is classified as fake (1) or not (0). For output convergence, the sigmoid activation function $\sigma(g)$ (Eq. 11) is used. The decision rule (to correspond the resultant vector with the classification label) is as expressed in Eq. 12, where g_{in} is the net input vector for the review.

$$\sigma(g) = \left(\frac{1}{(1 + e^{-g_{in}})} \right) \quad (11)$$

$$\sigma(g) = \begin{cases} 1, & \text{if } \sigma(g) > 0.5. \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

4 Experiments and evaluations

In this section, we present experimental setups and empirically evaluate the efficiency of the proposed model.

Datasets Amazon product reviews,¹ YelpCHI² collected by Rayana, S. and Akoglu, L. (2015) [59], and OSF fake review dataset³ contributed by Salminen et al. (2022) [9] are the three publicly available datasets used to evaluate performance. The selected datasets cover both balanced and imbalanced aspects. Amazon and OSF are equally distributed (balanced), while YelpCHI is an imbalanced dataset. Generally, they are labeled datasets with features including, but not limited to, overall rating, review text (plain text), verified purchases, different product categories, product IDs, user information, product, and review titles. The YelpCHI dataset contains hotel and restaurant reviews from Chicago. The OSF fake review dataset contains an authentic/ original (OR) review, presumably created by humans, and fake/ computer-generated (CG) reviews. Table 1 contains the dataset’s statistical summary.

Experiment Before actual experiments, we adopted several commonly used preprocessing techniques, including but not limited to normalization, tokenization of the review text, lemmatization, and stopwords⁴ removal. Our experiment was conducted in the Python language. We designed, trained, and evaluated our neural networks using TensorFlow [60] and Keras [61]. We split the dataset into training and testing in the ratio of 80:20, respectively. Training and validation sets in the ratio of 70:10, respectively [3]. The training hyperparameters are finetuned through the grid search strategy, and the optimal values are dropout=0.3, batch size=16, epochs=50, learning rate=3e-5, three (3) convolutional layers used (each CNN module used 2*2 filters).

Evaluation To evaluate DHMFRD –TER performance, we utilized the standard metrics measures, namely accuracy (*acc*), recall (*rc*), precision (*pr*), and F1 score (*f_l*). High precision indicates a low false positive (*F_p*) rate, high recall means a low false negative (*F_N*) rate, and a high F1 score value suggests that the model performs well in precision and recall [3] (Table 2). Figures 2, 3, and 4 depict the proposed model performance learning curves on different datasets.

Baselines To illustrate the effectiveness of the proposed DHMFRD–TER, we compared our model performance with several fake review detection state-of-the-art baselines. Generally, results clearly demonstrate that our proposed integrated model attains the best performance on all three datasets. Table 3 summarizes the experimental results for all the compared state-of-the-art baseline methods in terms of *acc*, *rc*, *pr*, and *f_l*.

¹ <https://www.kaggle.com/lievgarcia/amazon-reviews>

² <http://odds.cs.stonybrook.edu/yelpchi-dataset/>

³ <https://osf.io/tyue9/>

⁴ <https://code.google.com/archive/p/stop-words/>

Table 1 Datasets statistical summary

Datasets	Number of fake reviews	Number of non-fake reviews	Total
Amazon	10,500	10,500	21,000
YelpCHI	8,919	58,476	67,395
OSF	20,000	20,000	40,000

- **SWNN** [62]: Presented a NN(s) based model to learn review sentence composition in a document representation; special attention was paid to each sentence-weight computation.
- **CNN** [33]: They used CNN in integrating review word embedding vectors (n-gram features) and emotion features.
- **DFNN** [33]: They used DFFNN in integrating review word embedding vectors (n-gram features) and emotion features.
- **CNN-LSTM** [63]: They precisely used the combination of CNN and LSTM models with aspect-based sentiment analysis to classify reviews.
- **ACB** [6]: The review sentence's word importance is calculated using attention mechanisms and high-level n-gram features extracted using CNN. Afterward, the obtained vectors from each review are combined using Bi-LSTM.

Statistical test We employed a nonparametric Friedman test to compare accuracies with its variants [64, 65]. It is a well-known statistical test widely used in existing works. Friedman test is based on ranking r_j^i of each tested j^{th} method across i^{th} dataset, where $j \in \{1, 2, 3, \dots, k\}$ and $i \in \{1, 2, 3, \dots, n\}$. Ties in rank are resolved through the averaging of their rank position. Friedman test statistic with degrees of freedom $k - 1$ is expressed as in Eq. 13 below;

$$F_r = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1) \quad (13)$$

We computed ranks of accuracies in relation to the Friedman test, as shown in Table 4. The alpha α value for all the calculations is set to 0.05. Attending to the values of R_i , $k - 1 = 5$, $\chi_{\alpha, k-1}^2 = \chi_{0.05, 5}^2 = 11.07$, $F_r = 11.38$. This indicates there is significant differences in our solution versus baseline methods, in which the DHMFRD-TER and DFFNN ranked first and second, respectively.

Table 2 Confusion matrix

	Actual positive	Actual negative
Spam review predicted	T_P	F_P
Non-spam review predicted	F_N	T_N

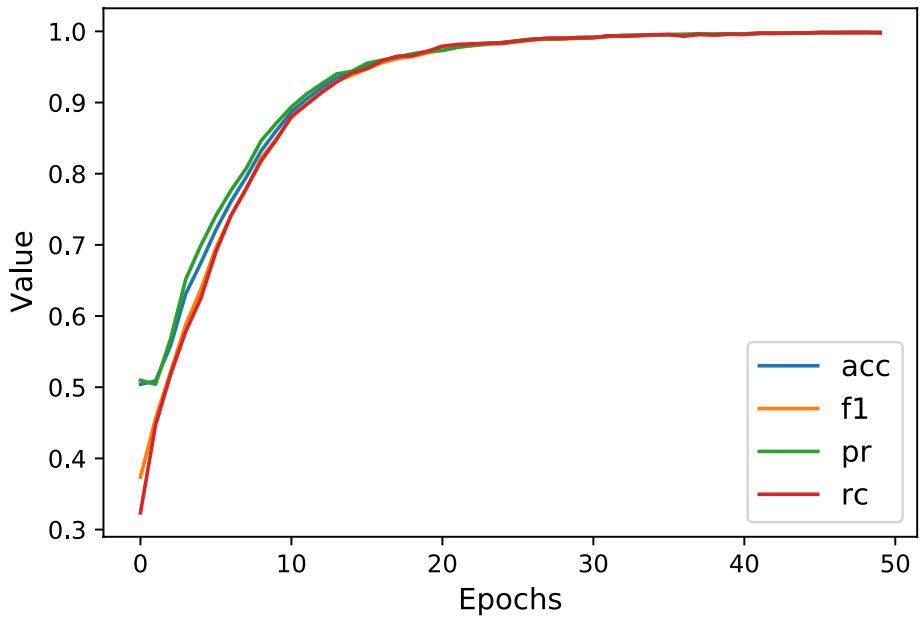


Fig. 2 Overall model performance learning curves for amazon products dataset

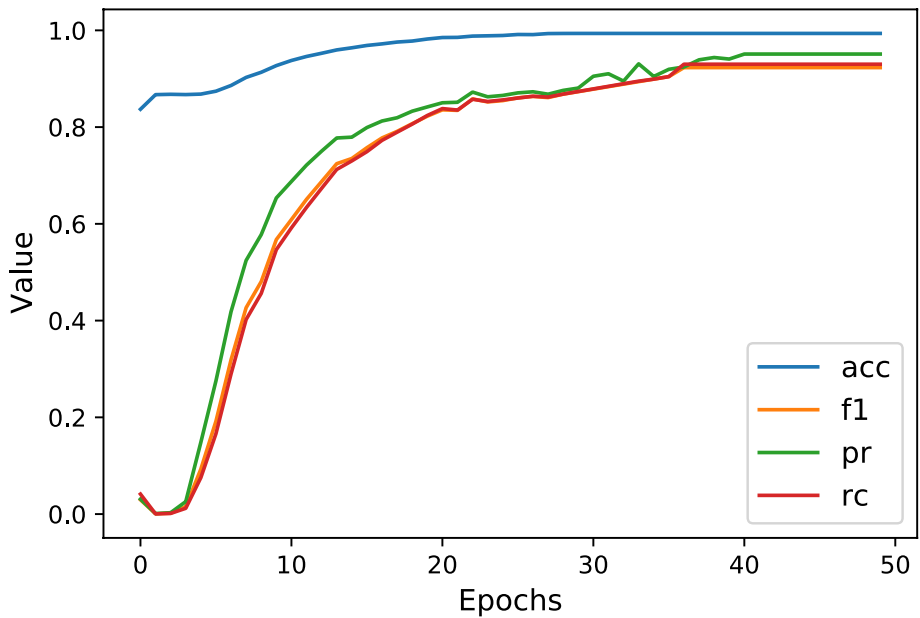


Fig. 3 Overall model performance learning curves for the Yelp CHI dataset

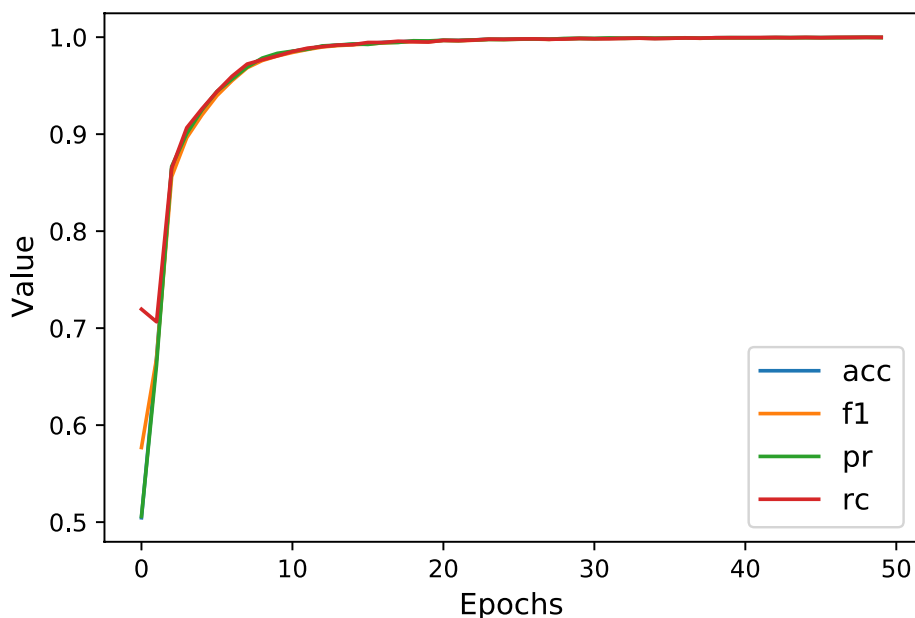


Fig. 4 Overall model performance learning curves for the OSF dataset

Furthermore, we observed lower performance results (Fig. 5) for all methods, including DHMFRD–TER on the imbalanced dataset (i.e., YelpCHI). This observation agrees with Budhi et al. (2021) [21] that class imbalance between fake and genuine reviews affects model performance.

5 Conclusion

The growth of online commerce is mutually linked to increased fraudulent reviews. The problem has prompted stakeholders and scholars to seek various solutions as they significantly impact customer’s purchase decisions. Most famous online business websites like Amazon.com and Yelp.com have mechanisms to prevent deceptive reviews. However,

Table 3 Comparison of experimental results on three datasets

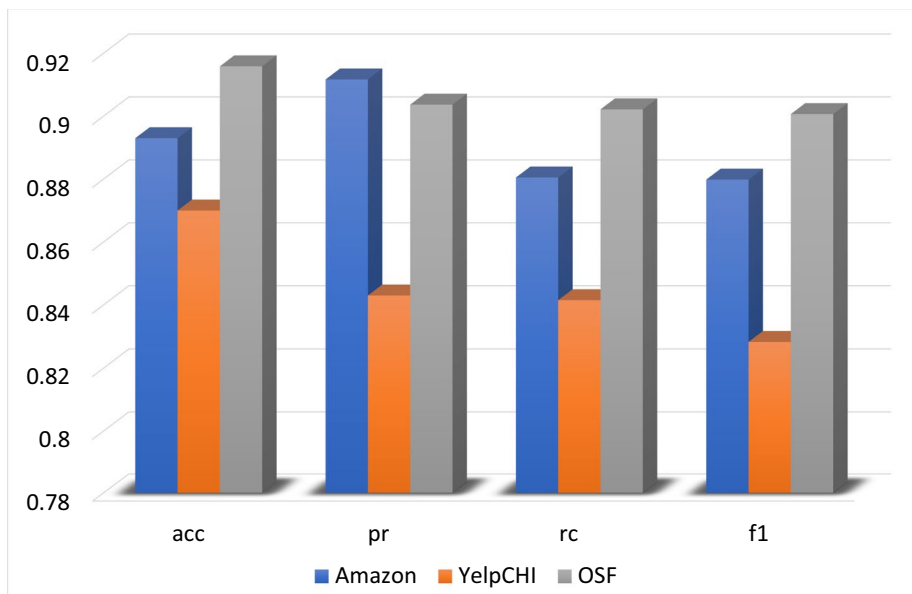
	Amazon dataset				YelpCHI dataset				OSF dataset			
	<i>acc</i>	<i>pr</i>	<i>rc</i>	<i>f₁</i>	<i>acc</i>	<i>pr</i>	<i>rc</i>	<i>f₁</i>	<i>acc</i>	<i>pr</i>	<i>rc</i>	<i>f₁</i>
SWNN	0.865	0.893	0.927	0.893	0.802	0.821	0.883	0.840	0.846	0.876	0.865	0.877
CNN	0.838	0.893	0.889	0.856	0.818	0.856	0.841	0.837	0.930	0.861	0.941	0.903
DFNN	0.898	0.905	0.899	0.921	0.875	0.859	0.864	0.872	0.927	0.912	0.923	0.929
CNN-LSTM	0.879	0.902	0.689	0.715	0.857	0.687	0.721	0.699	0.892	0.878	0.799	0.813
ACB	0.890	0.889	0.893	0.907	0.881	0.879	0.806	0.798	0.906	0.899	0.891	0.887
DHMFRD–TER	0.988	0.988	0.986	0.987	0.987	0.956	0.934	0.924	0.994	0.996	0.994	0.995

Table 4 Accuracies ranking for all datasets

SWNN	rank	CNN	rank	DFFNN	rank	CNN-LSTM	rank	ACB	rank	DHFRD-TER	rank
0.865	2	0.838	1	0.898	5	0.879	3	0.89	4	0.988	6
0.802	1	0.818	2	0.875	4	0.857	3	0.881	5	0.987	6
0.846	1	0.93	5	0.927	4	0.892	2	0.906	3	0.994	6
$\sum R_i$	4		8		13		8		12		18

spammers are still becoming very careful in avoiding being detected easily. Recently, spam detection model performance has been shown to improve by employing the DL over manual and basic ML approaches, as they can trigger the high dimensional and complex nature of reviews.

We proposed the DHFRD-TER in this study to detect fake reviews. To test the effectiveness of our model, we used three variable-sized datasets covering balanced and imbalanced dataset aspects in which results show DHFRD-TER performed better compared with state-of-the-art baselines. The DHFRD-TER achieved the best results in terms of accuracy (0.988), precision (0.988), recall (0.986), and f-measure (0.987) on the Amazon dataset; accuracy of (0.987), precision (0.956), recall (0.934), and f-measure (0.924) on the Yelp CHI dataset; accuracy of (0.994), precision (0.996), recall (0.994), and f-measure (0.995) on the OSF dataset. In addition, to demonstrate if there are significant differences in our solution versus baseline methods, we conducted Friedman statistical test to accuracies in which DHFRD-TER ranked first. The results of our experiment demonstrated that employing this kind of composite or high-dimensional feature maps is essential for surpassing existing techniques.

**Fig. 5** Average model performances on all the three datasets

Future work should have a deeper look at how to formalize the optimal number of features to be integrated, as stated by Luo et al. (2017) [66], that performance results are sacrificed slightly by not considering a certain feature during an experiment. Moreover, our extensive literature review has noted that the fake review detection arena also suffers from the unavailability of reliable labeled datasets. Most of the studies relied on Amazon Mechanical Turk (AMT) collected reviews or datasets collected using a similar collection technique of hiring Turkers to fake reviews. Even though these reviews are fake, they do not reflect real-world commercial website reality. Also, its legit is questionable, as it is difficult for all Turkers to have the same psychological state of mind as that of real fake reviewers with real business interests.

Funding This work was supported by the National Natural Science Foundation of China under Grant 62272048.

Declarations

Conflicts of Interest The authors declare that there is no conflict of interest in this Manuscript.

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Authors and Affiliations

Ramadhani Ally Duma^{1,2} · Zhendong Niu¹ · Ally Nyamawe² · Jude Tchaye-Kondi¹ · James Chambua³ · Abdulganiyu Abdu Yusuf¹

Ramadhani Ally Duma
radsiffi@gmail.com

Ally Nyamawe
ally.nyamawe@udom.ac.tz

Jude Tchaye-Kondi
tchaye59@gmail.com

James Chambua
jchambua@gmail.com

Abdulganiyu Abdu Yusuf
abdulg720@gmail.com

¹ School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China

² College of Informatics and Virtual Education, The University of Dodoma, Dodoma, Tanzania

³ Computer Science and Engineering Department, University of Dar-Es-Salaam, Dar-Es-Salaam, Tanzania